

Event Frequency Layer

Definition Document and Project Plan — MKM-EF-001

MKM Research Labs

Version 2.4 — 27-October-2026

David K Kelly, Chief Science Officer

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Contents

Status 4

1 What changed from version 1 5
- 1.1 What version 2.4 changes . . . 5
- 1.2 What version 2.3 changed . . . 5
- 1.3 What version 2.2 changed . . . 6
- 1.4 What version 2.1 changed . . . 6
- 1.5 What version 2.0 changed from version 1 . . . 7

2 Background and problem statement 8
- 2.1 Current state . . . 8
- 2.2 The defect, with evidence . . . 8
- 2.3 What is already correct . . . 9
- 2.4 The pattern already exists in this repository . . . 9
- 2.5 Remedy . . . 9

3 Objectives 10

4 Scope 10
- 4.1 Locked decisions . . . 10
- 4.2 In scope . . . 11
- 4.3 Chain position . . . 12
- 4.4 Out of scope for this phase . . . 12

5 Functional specification 12
- 5.1 Package layout . . . 12
- 5.2 Configuration . . . 13
- 5.3 Event-definition alignment . . . 13
- 5.4 Frequency abstraction . . . 14
- 5.5 Calibration pipeline . . . 14
- 5.6 Persistence . . . 15
- 5.7 The event catalogue . . . 15
- 5.8 The Monte Carlo year simulation . . . 15
- 5.9 The closed form, and why it is a test rather than the answer . . . 16
- 5.10 The multi-year term structure . . . 17
- 5.11 Wiring . . . 17
- 5.12 Extension points . . . 17

6 The λ provenance problem 17

7 Measured results 18
- 7.1 Reproducibility — a pre-existing defect, now fixed . . . 18
- 7.2 Cost and accuracy of the year simulation . . . 19

7.3 Repricing on halong 19
7.4 Extraction against the existing calculation 20
7.5 The clean run 20
7.6 The wind leg, exercised at last 20
7.7 What has not been validated 21

8 The validation arm cannot validate lambda **21**

9 Non-functional requirements **22**

10 Project plan **22**
10.1 Stage acceptance 23

11 Migration landmines **24**

12 Governance and registration **26**

13 Risks and assumptions **26**

14 Glossary **27**

Status

Document type	Definition Document and Project Plan
Component	Event Frequency Model — `MKM-EF-001` (new)
Version	2.4 — supersedes v2.3 of 24 July 2026
Status	Stages 1a/1b/1c, 3 and 4 implemented and wired to pricing; Stages 2 and 5 remain
Owner	CSO, MKM Research Labs
Proposed tier	1 (Maximum)

This document proposes a new model for the inventory and a staged plan to remove a pricing defect in the existing flood chain. The layer is built and now feeds pricing on the gauge, property, commercial and wind legs; Section 7 reports what it measures and, in Section 7.7, what it does not.

Two of the three defects corrected during this work were *pre-existing* and independent of the frequency model. One of them — an unseeded random draw that made every hazard curve irreproducible at roughly ±40% — is a governance matter in its own right and is set out in Section 7.1.

1 What changed from version 1

1.1 What version 2.4 changes

#	Change	Driver
C20	**The multi-year term structure is exact** (Section 5.10). Each year contributes its exceedance *intensity* $\lambda_y p$, so a tenor of T years carries $1 - \exp(-p\sum_y \lambda_y)$	It compounded each year's annual *probability* $1 - e^{-\lambda_y p}$ as though it were an intensity, understating the probability of a flood within the tenor by 0.1–17%, worst at one year and high λp. The term structure feeds the gauge PRS pricer
C21	**Annual probabilities are evaluated as** $-\text{expm1}(-\lambda p)$	$1 - \exp(-x)$ cancels at small x, losing about $10^{-16}/x$ of relative precision: 10^{-10} at $\lambda p = 10^{-6}$
C22	**Benchmarked against Wolfram** (Section 5.9)	`MKM-EF-001_wolfram_benchmark`: annualisation, the tenor probability, compound-Poisson loss and the count families against Wolfram Engine and a 50-digit reference. Version 0.1 found C20 and C21

1.2 What version 2.3 changed

#	Change	Driver
C16	**The per-gauge validation arm is circular.** It cannot check λ, because the threshold search targets λ	The peaks-over-threshold target is set *from* the catchment rate, so the search finds whatever threshold delivers it and recovers the seed by construction. Halong returned 4.48–4.58 against a seed of 4.5. **Nothing now validates** λ
C17	**Stage 1c complete**: exceedance *events* alongside exceedance *days* in the statistics module; fitted rates and provenance persisted through the `database` seam	The day count was never a rate. Persistence closes the BCBS 239 traceability requirement
C18	**The wind threshold is traced and now per-asset.** 55.56 m/s was a deliberate uniform constant (200 km/h), not a unit bug; `DesignWindSpeedKmh` drives the threshold instead, and design speeds were raised 40 km/h	Wind vulnerability was undifferentiated across a portfolio. L12 partially closes; the calibration question remains

#	Change	Driver
C19	**L13 resolved**: the peril stages record the BRI spine they read, and a guard warns when any stage under-declares	The warning fired after every successful run and had already misled one root-cause investigation. The gap was real: a change to the BRI spine never invalidated the consuming step

1.3 What version 2.2 changed

Wiring the layer into pricing and calibrating it on real halong data exposed three defects — two of them pre-existing and more serious than the problem this model was written to fix.

#	Change	Driver
C11	**The gauge response model was unseeded.** Fixed. Hazard curves were irreproducible at roughly ±40%; they are now byte-identical across rebuilds	Pre-existing, predating MKM-EF-001. Three unseeded uniform draws, two of which redrew each gauge's *character* on every build. The BCBS 239 reproducibility claim did not hold for this chain (Section 7.1)
C12	**Missing mild events were reweighted onto the severe end.** Fixed by scaling the conditional by catalogue *coverage* rather than renormalising the weights	The catalogue holds no minimal or baseline events, and those carry 73% of the population mass. A configured 8% severe-or-worse share became an effective 29.6% — a 3.7× overstatement
C13	**Stages 3 and 4 are built**: gauge, property, commercial and wind legs all price off the frequency layer	The reprice is **downward**, 0.35×–0.56× at the calibrated weights, not the upward move earlier drafts predicted
C15	**The wind leg is validated on real data**; L6 and L9 close	A `--typhoon` run put real wind through all ten commercial assets with the peril coherence checks passing. It also surfaced L12: the wind trigger is one global constant, not asset-differentiated
C14	**Two field-naming traps documented**, both of which produced confident wrong answers before being caught	`flood_events[].storm_id` holds *sequence* identifiers; `_load_gauge_hazard_curves` returns the *sequence* count while its caller names it `num_storms`

1.4 What version 2.1 changed

Version 2.0 was written before any code existed. Building it moved four things, one of which was an outright error in version 2.0's design.

#	Change	Driver
C7	λ **is a property of the catchment, not of a gauge.** Version 2.0 had per-gauge peaks-over-threshold calibration feeding λ into pricing. That is wrong: a per-gauge exceedance rate is *already* $\lambda \times P(\text{exceed} \mid \text{event})$, so using it as λ squares the conditional and double-counts. λ is now a seeded per-catchment rate, and per-gauge extraction became the **validation arm**	A storm arrives over the catchment and reaches every gauge in it. What differs per gauge is the conditional response, not the arrival rate
C8	**The Monte Carlo year simulation is the engine**, and has moved forward out of Track B. Version 2.0 deferred the sampler and priced off the closed form; the order is now reversed	The closed form is the simulation's *exact expectation*, not an approximation of it, so it is worth more as a check than as an answer. The simulation additionally yields the annual distribution, which the closed form cannot
C9	**Ten thousand simulated years**, and the reconciliation gate is expressed in sampling standard errors rather than as a fixed percentage	Measured on the target hardware. A fixed 2% band false-alarmed on 17% of runs at ten thousand years while never binding at a million
C10	**Landmine L3 resolved:** `num_storms` is kept and `num_events` added beside it	The storm/event distinction becomes visible in the data rather than hidden in a redefinition of a field existing consumers already read

1.5 What version 2.0 changed from version 1

Version 1 diagnosed the problem correctly. Version 2 keeps the diagnosis and reworks the plan against what the codebase actually does. Six substantive changes:

#	Change	Driver
C1	The layer is no longer greenfield — it is a **port of the existing seismic and fire occurrence pattern** onto the storm chain	`src/models/seismic/occurrence/` and `src/models/fire/initiation.py` already implement precisely the proposed $\lambda \rightarrow$ Poisson $\rightarrow$ event $\rightarrow$ severity construction. Storm and flood are the outlier, not the pioneer.
C2	Delivery splits into **Track A** (analytic annualisation) and **Track B** (ELT $\rightarrow$ YLT); only Track A is in scope	Version 1 went straight to the YLT sampler before any pricing benefit landed. Track A fixes pricing at one node without touching port generation.

#	Change	Driver
C3	Scope is a **generic per-peril frequency layer**, calibrated for storm first	The wind leg carries the identical defect; fire and seismic already carry their own λ. A flood-only layer would need a second frequency layer within months.
C4	New Section 6: λ **provenance is currently circular** on synthetic catchments	The "historical" gauge series is generated *from* the frequency field it would be used to calibrate. This materially changes what Stage 1 can claim.
C5	Version 1's "critical alignment task" is **already broken in a specific, locatable way** — sequences are shredded into storms before the denominator is taken	`src/models/hazard/io/_load.py:52`
C6	Non-functional requirements restated against the actual house rules and audit tooling	Version 1 cited `split_audit.py`, which no longer exists; specified 100% branch coverage against a house rule of $\geq$99% line; and did not mention the `database` seam at all.

2 Background and problem statement

2.1 Current state

The simulator generates synthetic storm events and evaluates, at each river gauge, whether a storm produces a flood. Insurance cost is priced as the ratio of flooding storms to total storms. This is a **conditional probability** — $P(\text{flood} \mid \text{storm})$ — and is dimensionless with respect to time.

Two gauges with identical conditional probability therefore receive identical prices even if one experiences eight qualifying storms a year and the other one. The true annualised risk differs by a factor of eight.

2.2 The defect, with evidence

The defect is not conceptual. It is written out in three places.

Location	Code	Meaning
`src/port/src/property/hc/pricing/_process.py:54`	`spread_bps = (flood_count / num_storms) * 10000`	The property PRS spread **is** $P(\text{flood} \mid \text{storm})$, expressed in basis points
`src/port/src/property/hc/pricing/_process.py:250`	`return_period_yrs = num_storms / flood_count`	The assumption stated explicitly: **one storm = one year**, $\lambda \equiv 1.0$

Location	Code	Meaning
`src/models/hazard/builder.py:109--141`	GEV fitted to the peak level of *every* simulated storm; the resulting exceedance probability assigned to `annual_exceedance_prob` and `annual_hazard_rate_*`	A per-event conditional is relabelled *annual* at the point of assignment

The same ratio construction appears on the gauge basis leg (`_process.py:119`) and the wind leg (`_process.py:70`, via `_wind_union`).

2.3 What is already correct

The downstream plumbing is rate-shaped and needs no rework:

- `src/models/hazard/gev.py:66` — `compute_term_structure` is already a Poisson compounding, $P(\geq 1 \text{ flood by year } t) = 1 - e^{-\lambda t}$.
- `src/models/hazard/prs_analytical.py:100` — already converts an annual probability to a continuous hazard rate via $-\log(1-p)$.

Both are being fed a per-event number. **The remedy is a substitution at one node, not a pricer rewrite.**

2.4 The pattern already exists in this repository

Two production models already do what version 1 proposed to invent:

- `src/models/seismic/occurrence/` — `_rates.py` computes a per-asset annual λ; `_simulate.py:89` draws `rng.poisson(lambda_annual * horizon_years, size=n_sim)` and instantiates events where the count is at least one; severity and damage follow.
- `src/models/fire/initiation.py` — the same construction, labelled "Model A — Poisson initiation".

MKM-EF-001 should mirror this package shape rather than introduce a parallel abstraction with different conventions.

2.5 Remedy

$$\text{Annual flood frequency}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \tag{1}$$

$$\text{Annual exceedance probability}_g = 1 - \exp(-\lambda_g \, P(\text{flood} \mid \text{event}, g)) \tag{2}$$

$$\text{Pure premium}_g = \lambda_g \times P(\text{flood} \mid \text{event}, g) \times \mathbb{E}[\text{loss} \mid \text{flood}, g] \tag{3}$$

with λ_g the calibrated annual **event** arrival rate — where "event" means the insurance hours-clause event, not the individual storm (Section 5.3).

3 Objectives

#	Objective	Success measure
O1	Price reflects event frequency	AAL differs proportionally between high- and low-frequency gauges with equal conditional flood probability
O2	Honest return periods	`return_period_years` derived from λ and the conditional, not from the storm count; the 100-year clamp reviewed
O3	Independently validatable components	Frequency backtestable separately from vulnerability; MKM-GH-001 outputs renamed so nothing claims to be annual that is not
O4	Compliance by construction	λ_g provenance recorded and reproducible (BCBS 239), *including honest recording of generator-derived provenance*; validation artefacts auto-generated (SR 11-7)
O5	No new technical debt	House rules R1–R6 satisfied; calibration persisted through the `database` seam, not JSON
O6	Quantified repricing before switchover	Per-gauge and per-property parallel-run delta report produced *before* the legacy metric is retired

4 Scope

4.1 Locked decisions

Decision	Choice	Rationale
Pricing engine	**Monte Carlo year simulation.** Draw how many qualifying events arrive in a year, ask of each whether it floods, repeat. The annual flood probability is the fraction of simulated years that flooded	Retains the platform's simulation framework rather than replacing it with a formula, and yields the whole annual distribution — occurrence and aggregate views, and return periods — not just a mean
Closed form	**Reconciliation self-test, not an alternative answer**	It is the simulation's exact expectation. Any disagreement beyond sampling error means one of the two is wrong, which makes it a real check rather than a restatement

Decision	Choice	Rationale
λ granularity	**Per catchment**, not per gauge	A per-gauge exceedance rate is already $\lambda \times P(\text{exceed} \mid \text{event})$; using it as λ would square the conditional
λ source	**Configured per-catchment seed** (Thames 4.5 events per year), with per-gauge extraction as the validation arm	Per house convention for a new probabilistic model. The data-driven estimator is blocked by Section 6 on synthetic catchments in any case
Peril scope	**Generic abstraction with per-peril** λ; calibrate storm first	Wind carries the same defect; fire and seismic already have λ and fold in later
Event aggregation	**Maximum level across the storms of a sequence**	A PRS pays on a level being breached, and a week containing two breaches is one breach of the contract, not two
Denominator	**Keep `num_storms`, add `num_events`**	Existing consumers read `num_storms`; redefining it in place would change their meaning silently
Chain position	**Downstream of storm severity**	Nothing upstream changes; one switch point
Model ID	`MKM-EF-001` — *Event Frequency Model*	`EF` is unused in the model inventory
Tier	**1** (proposed)	λ is a linear multiplier on every PRS spread — more material than MKM-SI-001 (tier 2, "not directly into pricing"). Peers at tier 1: MKM-GH-001, MKM-PR-001, MKM-DE-001

4.2 In scope

- Frequency abstraction with Poisson and Negative Binomial implementations, per peril.
- Calibration pipeline: peaks-over-threshold extraction, declustering, arrival-rate estimation, dispersion diagnostics, per-gauge model selection.
- **Event-definition alignment**: the hours-clause event (a storm *sequence*), not the individual storm, becomes the unit of both λ and the conditional.
- The annualisation seam and its wiring into MKM-GH-001.
- Repricing of the property and gauge PRS legs.
- Calibration persistence and provenance through the `database` seam.
- Validation module: dispersion test, count backtest, threshold sensitivity, reconciliation, POT round-trip.
- Configuration schema extension in a new `config/frequency/` package.

- Governance pack: inventory entry, model documentation, registry wiring.

4.3 Chain position

MKM-SI-001 (severity)

↓

MKM-SG-001 (gauge response)

↓

MKM-GH-001 (per-event exceedance)

↓

MKM-EF-001 — catchment λ, year simulation

↓

MKM-PR-001 (PRS pricing)

Physically, frequency is a *sibling* of severity rather than a descendant. As an implementation seam, downstream is the right choice: nothing upstream changes, MKM-GH-001 is merely **relabelled** — its outputs become honest per-event conditionals — and `builder.py` is the single switch point.

Inventory `upstream_models` and `downstream_models` must be updated on MKM-GH-001 and MKM-PR-001 accordingly; chain consistency is scanned by full-audit section 4.7.

4.4 Out of scope for this phase

- **Track B**: ELT export, YLT sampler, AEP construction, portfolio aggregation, cross-gauge count correlation.
- Non-stationary or climate-conditioned $\lambda(t)$. Architected for, not implemented.
- Seasonal and non-homogeneous Poisson rates.
- Wind, fire and seismic λ *calibration* — the abstraction accommodates them; only storm is calibrated in this phase.
- Changes to the inundation model, the spatial model or the DEM integration.
- Multi-year contract pricing and discounting.

5 Functional specification

5.1 Package layout

Mirrors `src/models/seismic/occurrence/`. Every file under 300 lines (R2); no callable code in any `__init__.py` (R4); canonical copyright header on every file (R5).

```
src/models/frequency/
  __init__.py             *  module docstring + re-exports only
  annualise.py            *  the seam: P = 1 - exp(-lambda * p_event)
  calibrate.py            *  pure function of (series, config) -> FittedRate
  datastructures/
    _rate.py              *  FittedRate: lambda, threshold, diagnostics
    _provenance.py        *  CalibrationProvenance, ProvenanceClass
    _catalogue.py         *  EventCatalogue: events x gauges, + coverage
```

```
    _frame.py              *  EventFrame: grouping without gauge levels
    _simulation.py         *  EventDraws, YearSimulation
  pot/
    _threshold.py          *  rate-targeted threshold search
    _decluster.py          *  runs-method independent peaks
    _extract.py            *  series -> declustered peaks -> annual counts
  events/
    _identity.py           *  which event a storm belongs to
    _aggregate.py          *  per-storm responses -> catalogue / frame
    _weights.py            *  population weights + catalogue coverage
  ylt/
    _sample.py             *  draw_event_years / apply_catalogue
    _reconcile.py          *  closed form + sampling-error gate
  families/
    _poisson.py               Stage 2
    _negbin.py                Stage 2
    _select.py                Stage 2 -- dispersion test, logs overrides
```

Listing 1: Package layout. Items marked * are built and under test.

`annualise.py` is deliberately tiny and deliberately the only place the composition happens. Frequency logic must not leak into the event simulator or the pricing modules.

The dependency runs **hazard** → **frequency** → **config**, one way. Event identity lives in `events/_identity.py` rather than beside the loader that writes the tag, because siting it with the loader would have closed an import cycle the moment the hazard builder began consuming the frequency layer.

5.2 Configuration

A new `config/frequency/` package with `_schema.py` and `_loader.py`, following the hydrated-dataclass pattern of `config/seismic/`. It owns, per house rule R1:

- Target mean exceedance rate band for threshold selection.
- Declustering window (minimum inter-event separation).
- Dispersion-test critical value; AIC tiebreak policy.
- Minimum record length for per-gauge calibration; regional fallback λ.
- Return-period grid for output tables.
- Per-peril family overrides, with the override itself logged (SR 11-7).

It also absorbs an existing orphan: `src/models/intensity/distribution/_core.py:198` carries `storms_per_year: float = 20` as a default argument — an undocumented frequency assumption living inside the severity model, and an R1 breach. That parameter moves here and the severity model takes it as an injected value.

5.3 Event-definition alignment

This is not a hypothetical risk. It is already wrong, and locatably so.

MKM-SS-001 produces **sequences** — one to five related storms fitted inside a 168-hour insurance hours clause (`EVENT_WINDOW_HOURS` in `config/port/_storm.py`). But `src/models/hazard/io/_load.py:52--61` flattens every sequence into its member storms before the hazard model sees them.

`num_storms` is therefore one to five times the event count, and today's denominator is **neither events nor years**.

Built. The remedy as delivered:

1. `load_storms_from_sequences` tags each storm with its `sequence_id`. The key is **added, not substituted**, so every existing consumer of the storm fields is untouched. An untagged sequence falls back to a positional identity, so two untagged sequences cannot silently merge into one event.
2. `event_id` and `count_events` live with the loader that writes the tag, and the frequency aggregator imports them, so the fallback rule cannot drift between the two. This also keeps the dependency pointing downhill: frequency imports hazard, never the reverse.
3. The conditional is evaluated per event — the **maximum** peak level across the member storms, not one row per storm.
4. Grouping is by storm **identity**, not position. The two coincide today, but position would break silently if response ordering ever changed.
5. λ is quoted in **hours-clause events per year**, matching that unit exactly.
6. `num_storms` is kept and `num_events` added beside it in both the hazard metadata and the build summary.

The physics is untouched: each storm is still routed through the gauge response model individually. Only the grouping changes.

This step reprices on its own, before λ is applied, because the denominator changes. It was sequenced first precisely so its effect could be measured apart from the λ effect — and doing so was worthwhile, because the two turned out to push the *same* way rather than offsetting (Section 7).

5.4 Frequency abstraction

A single interface, per peril, following the seismic rates/simulate split:

- `fit(counts: AnnualCountSeries) -> FittedRate`
- `annual_rate() -> float`
- `sample_annual_count(rng) -> int` *(unused in Track A; required by Track B)*
- `diagnostics() -> FrequencyDiagnostics`

Implementation	Distribution	Selected when
`PoissonFrequency`	Poisson(λ)	Dispersion index ≈ 1 (default)
`NegativeBinomialFrequency`	NegBin(r, p)	Overdispersed counts (variance $>$ mean)

Selection is data-driven per gauge via the dispersion index with an AIC tiebreak, recorded in the calibration output. A configuration override may force a family; the override is logged with its justification.

5.5 Calibration pipeline

Per gauge, from the gauge level record:

1. **Threshold selection** — the threshold yielding a target mean exceedance rate, anchored on bankfull discharge where available. Diagnostics: mean-excess and threshold-stability.

2. **Declustering** — enforce a minimum inter-event separation so peaks are independent. Record the rule applied.
3. **Arrival rate** — $\hat{\lambda}$ = exceedances / years of record; the annual count series is retained for dispersion testing.
4. **Dispersion test and family selection**.
5. **Severity cross-check (validation only)** — fit a GPD to exceedance magnitudes and compare against the simulator's implied severity at that gauge. Divergence beyond tolerance raises a validation flag; it never silently recalibrates anything.

Prior art to fix, not to write from scratch. `src/models/statistics/timeseries.py:103--122` already computes a per-gauge `frequency_per_year` for alert, warning and severe. It counts **days over threshold**, not declustered events — peaks-over-threshold with step 2 missing — and therefore overstates λ by roughly the mean event duration in days. Stage 1 is a fix-and-promote of this calculation into the new package, with the existing consumers repointed.

Provenance (BCBS 239). Every fitted rate carries: source dataset identifier and version, record period, threshold, declustering rule, family selected and why, fit timestamp, code version, configuration hash, and a **provenance class** — `observed`, `generator-derived` or `regional-fallback`. Calibration is a pure function of (data, configuration): re-running with identical inputs must reproduce identical outputs, regression-tested against a golden master.

5.6 Persistence

Fitted rates and their provenance are written and read **exclusively through the `src/database/` seam** (R6). No JSON artefact is created at any point — full-audit section 4.5 is walking JSON writes toward zero tolerance, and a new model must be born database-native.

Deliverables: a new table plus Alembic migration following the existing gauge hazard-curve migration, with matching entries in the ORM model and column maps, and accessor functions in the `database` package API.

5.7 The event catalogue

One row per event, one column per gauge, holding the highest level that event drove at that gauge. This is what the year simulation resamples from.

Because every gauge's outcome for a given event sits in the **same row**, drawing an event row draws a spatially coherent storm. The correlation between gauges is carried by the catalogue itself and does not have to be modelled again downstream.

The catalogue also exposes the conditional half of the decomposition directly: $P(\text{flood at } g \mid \text{event})$ is the fraction of catalogue rows whose level at g clears the trigger.

5.8 The Monte Carlo year simulation

For each simulated year: draw $N \sim \text{Poisson}(\lambda)$ qualifying events, resample N catalogue rows with replacement, and record how many of them flood. The annual flood probability is the fraction of years with at least one flood.

The draw is separated from its application. `draw_event_years` runs once per run; `apply_catalogue` runs per subject against those same draws. A storm arrives over the catchment and reaches every gauge and property in it, so one set of draws must serve every subject. Drawing independently per subject would discard the spatial correlation the catalogue supplies: measured, two gauges on one reach correlate **0.78** under shared draws and −0.004 under independent ones, so a two-hundred-gauge book would look two hundred times better diversified than it is.

Resampling, not regeneration. The catalogue's outcomes are already computed, so a simulated year is index lookups rather than a re-run of the hydrology. That is what makes the year count affordable.

Outputs, all from the same run:

- **Annual flood probability** — the occurrence view; what the PRS spread is priced from.
- **Expected floods per year** — the aggregate view. It exceeds the occurrence probability whenever a year can carry more than one flood, which is exactly the case a conditional-only model cannot represent at all.
- **Return-period levels** — quantiles of the annual distribution.
- **Event exceedance frequency** — valid below the one-year return period, which matters at frequent-flood gauges and cannot be expressed as a probability.

All outputs carry the frequency-model identifier and calibration provenance in their metadata.

5.9 The closed form, and why it is a test rather than the answer

For $N \sim \text{Poisson}(\lambda)$ with each event flooding independently with probability p:

$$\mathbb{E}\big[(1-p)^N\big] = \exp\big(\lambda((1-p)-1)\big) = \exp(-\lambda p) \tag{4}$$

so $P(\text{at least one flood in a year}) = 1 - \exp(-\lambda p)$ **exactly**. The closed form is not an approximation of the simulation — it is the simulation's expectation. The same argument gives the aggregate view directly as λp.

Wolfram Engine proves the identity symbolically, and with a 50-digit reference confirms the code computes it to better than 10^{-12} (`MKM-EF-001_wolfram_benchmark`, filed on the model under Benchmark).

That is precisely what makes it valuable as a check: any gap beyond sampling error means one of the two is wrong. It is wired in as a reconciliation gate, not as the production pricing path, because the simulation additionally yields the whole annual distribution.

The gate measures the gap in **sampling standard errors**, $\sqrt{p(1-p)/n}$, against a configured band, rather than as a fixed percentage. A fixed band cannot mean the same thing at every year count: at ten thousand years a 2% band false-alarmed on 17% of runs, while at a million it never bound at all. The statistic is verified to be a true z-score — mean $|z|$ measures 0.811 and 0.797 at ten and a hundred thousand years against the theoretical $\sqrt{2/\pi} = 0.798$ — and that calibration is itself a test, so the band cannot silently stop meaning what it says.

5.10 The multi-year term structure

A PRS runs for several years, and the arrival rate may move across them: a *rate process* gives λ_y for contract year y, constant (`ConstantRate`) or growing geometrically (`TrendRate`). Exceedances in year y are Poisson with intensity $\lambda_y p$, so survival over t years is the product of the yearly survivals and

$$P(\text{at least one flood by year } t) = 1 - \exp\Big(-p \sum_{y=0}^{t-1} \lambda_y\Big). \tag{5}$$

`annual_hazard_by_year` supplies the intensities $\lambda_y p$ and `compute_term_structure` accumulates them, so every point of the term structure equals `term_exceedance_probability` for its tenor, and its expected flood count is $p \sum_y \lambda_y$. Each probability of the form $1 - e^{-x}$ is evaluated as $-\text{expm1}(-x)$, which keeps full precision when x is small.

5.11 Wiring

#	File	Change	State
1	`src/models/hazard/io/_load.py`	Sequence tagging; `event_id` and `count_events`; `num_events` in metadata and summary	**Built**
2	`src/models/hazard/builder.py:109--141`	GEV fits per-event peaks; outputs renamed to `event_exceedance_prob`; annual rates come from the simulation	Stage 3
3	`src/port/src/property/hc/pricing/process.py:54,119,250`	Pricing spread, gauge basis leg and return period become simulation-derived	Stage 4

5.12 Extension points

λ as a process rather than a constant is now built: constant and trending rate processes drive the multi-year term structure (Section 5.10). Architected now, built later: a per-peril λ registry so wind, fire and seismic attach without restructuring; and, because the simulation scores a boolean flood outcome per event, replacing that boolean with a loss quantum turns the same machinery into a full year loss table without changing the sampler.

6 The λ provenance problem

Finding. On synthetic catchments, calibrating λ from the gauge "historical" record is circular.

`src/models/statistics/synthetic.py:122--124` generates the daily series by injecting severe exceedances at exactly the rate implied by the gauge's `FrequencyExceedLevel3` field:

```
severe_floods_expected    = max(freq_exceed_level3, 1)
severe_flood_prob_per_day = severe_floods_expected / (years * 365)
```

and `FrequencyExceedLevel3` is itself produced by `src/port/rand/shared/gauge/gauge_field_generators.py:168--170` as a `random.choices(...)` draw over a five-year window.

A peaks-over-threshold extraction over that series therefore recovers the injected rate. **The only per-gauge frequency number the platform currently holds is a random integer.**

Consequences.

1. Version 1's stated risk — "short or gappy records give a noisy $\hat{\lambda}$" — is the wrong risk. The real one is that λ has no observational content on synthetic catchments.
2. It does not invalidate the POT pipeline. POT is exactly what would be run against real gauge series, and building it now is correct.
3. It does change what Stage 1 may claim. The provenance class must record `generator-derived`, not `observed`, and the validation report must say so.

Approach adopted.

- **Now:** build POT as specified. On synthetic catchments, validate it by **round-trip** — the extracted $\hat{\lambda}$ must recover the injected rate within tolerance once declustering is accounted for. This is a genuinely strong test of the extraction code, and it is labelled as such rather than presented as calibration.
- **Follow-on** (separate approval; requires port regeneration): make `FrequencyExceedLevel3` a *modelled* field rather than a random one, giving the gauge generator a catchment-level λ prior so synthetic frequencies are physically plausible and spatially coherent. Gauges on the same reach should not carry independent λ.

7 Measured results

Everything here is measured on the halong catchment (3 gauges, 1 residential property, 10 commercial assets, 1000 storm sequences) unless stated otherwise. It supersedes every estimate in earlier versions.

7.1 Reproducibility — a pre-existing defect, now fixed

`GaugeResponseModel` drew from three **unseeded** uniform random calls. Two of them redrew each gauge's *fundamental character* — its normal level and response coefficient — on every build. Twelve rebuilds of one gauge, identical inputs:

Metric	Mean	Min	Max	Spread
Frequency-derived annual probability	0.1821	0.1332	0.2201	**48% of mean**
Legacy annual probability	0.1168	0.0923	0.1341	**36% of mean**

This predates MKM-EF-001 entirely. **Every hazard curve the platform has ever produced was irreproducible at roughly ±40%**, and the BCBS 239 reproducibility claim did not hold for this chain. It is not a frequency-model problem; it was found because calibrating the frequency model required repeatable measurements.

The fix seeds per gauge and per (gauge, storm), hashing the identifiers with a configured run seed rather than drawing from one global stream. A global seed would have been worse than useless: every gauge's character would shift whenever an unrelated gauge joined or left the portfolio. Verified —

identical across rebuilds, unchanged when another gauge is removed, and hazard curves now hash byte-identical between consecutive builds.

This is a governance event in its own right. Re-running any existing catchment now gives a different — and from now on stable — answer, independent of anything MKM-EF-001 does.

7.2 Cost and accuracy of the year simulation

Apple M2, 8 cores, 8 GB. A full portfolio is 2,200 subjects scored against one shared set of draws.

Simulated years	Full portfolio	Sampling error
1,000	0.07 s	∼5%
10,000 *(configured)*	**0.40 s**	**∼1.6%**
100,000	4.1 s	∼0.5%
1,000,000	43 s	∼0.16%

Run-to-run stability does not depend on this number — the seed is pinned — so what the year count buys is closeness to the true expectation, not repeatability.

7.3 Repricing on halong

Catalogue: 2112 storms to **1000 events** (2.11 storms per event). Event category mix 421 moderate, 336 severe, 197 extreme, 46 catastrophic — **no minimal or baseline at all**, confirming the importance-weighting is load-bearing rather than theoretical. Coverage 0.27. $\lambda = 4.5$ events per year. Severe-or-worse population share calibrated to 8% by the model owner.

Gauge	$P(\text{fl} \mid \text{ev})$	New	Old	Ratio	New RP	Old RP
GAUGE-419b4a25	0.0145	0.0632	0.1120	0.56×	15.3y	8.9y
GAUGE-635036ec	0.0124	0.0542	0.1272	0.43×	17.9y	7.9y
GAUGE-789592e7	0.0125	0.0546	0.1022	0.53×	17.8y	9.8y
SYNTH-182600fa	0.0100	0.0440	0.1266	0.35×	22.2y	7.9y

Property leg: 1100 bps to 587 bps (0.53×), return period 9.1y to 16.5y. **Commercial leg:** all ten assets priced, 6.8 to 518.1 bps, with inclusion-exclusion and union/intersection ordering verified across the peril legs.

The reprice is therefore **downward, 0.35×–0.56×**, not the upward move every earlier draft of this document predicted. The direction reversed twice during development — first when population weighting was added, again when the coverage treatment was corrected — which is the strongest argument in this document for not quoting a repricing figure until the mechanism beneath it is settled.

Severe-flood return periods of 15–22 years against the legacy model's 8–10 may still be too benign for a typhoon-exposed catchment. The lever is the event-population weighting; it now behaves linearly and reproducibly, and a rebuild takes about ten seconds.

7.4 Extraction against the existing calculation

On a thirty-year synthetic gauge record, at the severe trigger, the existing statistics module reports **1.234** exceedance *days* per year against **0.967** declustered *events* — a **28% overstatement**, being the mean flood duration in days. The fix-and-promote that removes it is still outstanding.

7.5 The clean run

A full port with the frequency layer wired throughout (halong, 3 gauges, 1 property, 10 commercial assets, 1000 sequences, 23 July 2026):

Check	Result
Hazard curves reproducible	**Identical** across rebuilds — the seeding fix holds outside the test harness
Gauge severe return period	**15.8–17.8 years** at $\lambda = 4.5$ and the 8% share
Property PRS	**587.2 bps**, return period 16.5 years
Commercial PRS	10 assets, 75–606 bps
Wind leg	Zero everywhere — see below

The run also caught a defect no test had: **flood transmission rates of 183%, 196% and 200%**, against a quantity bounded by 1 by construction. The basis leg derived its gauge count as annual probability times scenario count — sound while that probability was a per-event conditional, and a units error the moment Stage 3 annualised it. The hazard builder now writes a raw catalogue count for the basis leg, and the rates return to 0.43–0.51. The headline spread is unaffected.

Two things are worth drawing out. The error was found by a *summary line in a run log*, not by 2800 passing tests — the invariant that would have caught it is asserted nowhere. And it is the same class of units error this document already records twice, committed this time by the author of those warnings.

7.6 The wind leg, exercised at last

A second run with `--typhoon` generated 1000 coupled typhoons and put real wind through the peril legs for the first time.

Check	Result
Commercial assets with a non-zero wind leg	**10 of 10**
Inclusion-exclusion; union $\geq$ both legs; intersection $\leq$ either	**PASS**
Hazard curves reproducible across rebuilds	**Identical**
Flood transmission rate	**55.7%** property, **29.7%** commercial

The peril fan is live throughout — property `win` 295.6, `faw` 151.8, `fow` 760.8 bps; commercial `win` 31.9, `faw` 24.2, `fow` 372.0 bps. **L6 and L9 are both closed**, and the transmission rates confirm the units fix in a fresh run rather than in a patched rebuild.

The wind trigger is not asset-differentiated. All ten commercial assets trigger on the same five events. The wind field *is* sampled per location — `peak_sustained_ms` differs per asset — but only in the fourth decimal, the portfolio being tightly clustered. The deciding quantity, `v_50_eff_ms`, is **55.56 for every asset**: a Hotel, an Office, three MultiFamily and two Retail all share one vulnerability threshold. And 55.56 m/s is exactly 200 km/h ($200 \div 3.6 = 55.5556$), a round design figure in one unit surfacing as a threshold in another. Wind exposure is therefore driven by a single global constant rather than by construction type. This sits in MKM-WD-001's territory, not this model's — the annualisation is correct given whatever trigger it is handed — but the wind spreads above describe one threshold, not asset-level resilience.

Run cost. The typhoon stage took 15 minutes for 1000 events at 100 particles, and the ensemble write a further ∼20 on a machine already deep into swap. A run that appears hung after the progress bar reaches 100% is most likely still serialising.

7.7 What has not been validated

- **Wind vulnerability is not asset-differentiated** (L12). The wind leg now runs on real data and its arithmetic is verified, but the trigger it consumes is one global constant, so the wind spreads describe a threshold rather than a portfolio.
- **No full-book parallel run.** Everything measured here is a single small catchment: 3 gauges, 1 property, 10 commercial assets.
- λ **itself remains a seed**, and on synthetic catchments cannot be calibrated from the gauge record at all.

8 The validation arm cannot validate lambda

Version 2.1 demoted per-gauge extraction from *producing* λ to *validating* it. Running that validation on halong in Stage 1c showed it cannot do the job either, for a reason independent of the synthetic-record circularity.

The peaks-over-threshold search target is set **from the catchment arrival rate** — deliberately, so the qualifying-event threshold and λ describe the same event population rather than drifting apart. But the search then picks whatever threshold delivers that rate, so the recovered rate equals the seed by construction:

Gauge	Recovered λ	Seed
GAUGE-419b4a25	4.523	4.5
GAUGE-635036ec	4.483	4.5
GAUGE-789592e7	4.583	4.5
SYNTH-74cf6170	4.503	4.5

A quantity that reproduces its own input is not a measurement. The arm validates the **extraction code**, not the rate. Combined with the synthetic-record problem, **nothing in the platform currently constrains** λ, and landmine L8 is correspondingly wider than earlier versions of this document implied.

Two things the arm can still say, and both are worth having:

- **Dispersion.** Halong's four gauges returned a dispersion index of **0.46–0.97** — annual counts *less* variable than Poisson. That bears directly on Stage 2: under-dispersion is fitted badly by Poisson and worse by Negative Binomial, which models only the over-dispersed side.
- **The decomposition identity.** $\lambda \times P(\text{exceed} \mid \text{event})$ reproducing a gauge's directly measured exceedance rate remains a genuine consistency check on the composition, independent of whether λ itself is right.

Breaking the circularity needs a threshold anchored outside the rate being tested — bankfull discharge, or a published national flood-frequency estimate — and therefore real gauge data.

9 Non-functional requirements

Restated against the project coding rules.

Rule	Requirement for this project
R1 — no parameters outside `config`	All thresholds, declustering windows, selection criteria, fallback λ and return-period grids live in `config/frequency/`. The orphan `storms_per_year=20` is absorbed.
R2 — no file over 300 lines	Package layout adopted from the outset.
R3 — ≥99% coverage, per stage	Verified at the end of *each* stage. Version 1 specified 100% branch coverage; aligning to the house rule avoids a bespoke standard for one package. Plus property-based tests for mean and variance recovery, and a golden-master calibration regression.
R4 — no functions in `__init__.py`	Re-exports only.
R5 — canonical copyright header	On every new source file; enforced by the self-healing header audit.
R6 — data store only via `database`	Zero direct SQL, ORM or connections; zero JSON artefacts.

Additionally: seeded RNG throughout, with calibration and simulation reproducible from (data version, configuration version, seed); a structured run log carrying inputs, configuration hash, code version, outputs and validation flags; and verification through the full audit — *not* `split_audit.py`, which version 1 cited and which no longer exists. Relevant full-audit sections: 4.3 path definitions, 4.5 JSON files, 4.6 database usage, 4.7 model chain consistency.

10 Project plan

Leaves first; each stage independently shippable and tested; coverage checked at every stage boundary.

The staging has changed from version 2.0. The year sampler was version 2.0's Stage 6 and is now built, because it became the engine rather than a deferred extension. Stage 1 is correspondingly larger

and partly done.

Stage	Content	State	Reprices?
1a	Event-definition alignment; event catalogue; peaks-over-threshold extraction; per-catchment λ; `num_events`	**Done**	Denominator only
1b	Monte Carlo year sampler; shared draws; closed-form reconciliation gate	**Done** *(was Stage 6)*	Not consumed directly
—	*Unplanned, forced by calibration:* seed the response model; catalogue coverage	**Done**	**Yes**
3	Frequency layer wired into the hazard builder; legacy metric retained alongside	**Done**	**Yes**
4	Property, commercial, gauge-basis and wind legs; return periods	**Done**	**Yes**
1c	Day-count fix-and-promote; λ and provenance persisted through the `database` seam	To do	No
2	Poisson and NegBin families; dispersion selection; round-trip validation	To do	No
5	Governance: inventory entry, model documentation, registry wiring, validation report, monitoring job	To do	No
6	*Separate approval:* loss-weighted year loss table, ELT export, wind λ calibration, seasonal rates	To do	—

The clamp review promised for Stage 4 — the exceedance floor in the hazard builder and the 100-year return-period cap — is **still outstanding**. Return periods now run 15–22 years, so neither clamp binds today, but they remain unreviewed and would bind at a lower severe-or-worse share.

10.1 Stage acceptance

- **S1a/S1b** — *met.* Event-granular loader covered; catalogue and sampler under test; the simulation reconciles with its closed form.
- **S3/S4** — *met, with one gap.* Gauge, property, commercial and wind legs price off the frequency layer; hazard curves are reproducible across rebuilds; the port suite passed 2812 against a 2807 baseline with zero regressions; the legacy metric is retained alongside for parallel run. **The wind leg has not been exercised on real data** — see Section 7.7.
- **S1c** — peaks-over-threshold series reproducibly generated for all gauges; provenance complete and persisted through the seam; the day-count consumers repointed.
- **S2** — families pass property tests for mean and variance recovery; calibration deterministic under fixed inputs; per-gauge selection report generated; round-trip recovers the injected rate within tolerance.

- **Full-book parallel run** — outstanding. Requires a port run with the peril steps regenerated (L9), which is also what would first exercise the wind leg.
- **S5** — validation report reviewed and signed off; monitoring scheduled; legacy metric formally deprecated.

11 Migration landmines

These bite between Stage 3 and Stage 5. Each needs a decision before Stage 3 lands.

#	Landmine	Detail	Status
L1	**Magnitude of the reprice**	Measured at **0.35×–0.56×** at the calibrated 8% share — *downward*. The direction reversed twice in development.	**Quantified, not signed off.** A full-book parallel run on regenerated data is the gating artefact.
L2	**Compensating clamps**	The exceedance floor and the 100-year return-period cap exist because return periods were fake.	**Open.** Neither binds at 15–22 years, but both remain unreviewed and would bind at a lower severe-or-worse share.
L3	**`num_storms` is persisted and rendered**	Read by the user interface, the blotter and end-of-day.	**Resolved.** Kept, with `num_events` beside it. No consumer changed.
L4	**End-of-day history spans the cut**	Stored hazard rates change semantics *and* magnitude.	**Open, and wider than first thought.** The response-model seeding moves every curve independently of MKM-EF-001, so the discontinuity is not attributable to the frequency model alone.
L5	**Per-storm flood-event collapse**	—	**Resolved, and the premise was wrong.** The property leg was already event-granular and already divided by the sequence count. Only λ was missing.
L6	**Wind leg diverges**	Wind divided by the scenario count while flood was annualised, making the union and intersection legs incoherent.	**Resolved in code, unvalidated on data.** Both legs annualise on one frame in sequence space; halong cannot exercise it (L9).

#	Landmine	Detail	Status
L7	**Correlation is load-bearing**	Portfolio risk depends on subjects sharing one set of event draws; a per-subject caller would silently decorrelate the book (0.78 versus −0.004).	**Open.** Guard at the portfolio entry point; review point for any new caller.
L8	**Population weights are load-bearing and unvalidated**	The conditional scales directly with them; set to an 8% severe-or-worse share on the owner's judgement.	**Open.** Nothing currently contradicts a wrong value; the per-gauge extraction arm is the intended check and depends on Stage 1c.
L12	**The wind trigger is a single global constant**	`v_50_eff_ms` is 55.56 m/s for every commercial asset regardless of construction type — exactly 200 km/h, a round figure in one unit appearing as a threshold in another. All ten assets consequently trigger on the same five events.	**Open, and outside this model.** MKM-WD-001 owns it. Any wind spread quoted from MKM-EF-001 inherits it.
L13	**The stale-lineage warning is unreliable**	`commercial_peril_ts` and `property_peril_ts` are reported stale after runs in which both demonstrably executed and produced triggers.	**Open.** A warning that cries wolf trains readers to ignore it, which is how the L9 root cause went misdiagnosed for a day.
L9	**The typhoon step is opt-in, so `--all` produces no wind**	Corrected from a clean run. The peril timeseries steps *do* regenerate; the fault is upstream. `--all` does not run typhoon, so damage records are whatever a previous run left — on halong, 25 asset identifiers with **zero overlap** against the 10 the run generated. The stale-lineage warning is a symptom, not the cause.	**Resolved.** A run with `--typhoon` put real wind through all ten assets and the coherence checks passed. The flag requirement stands as an operational note.
L11	**Pricing invariants are not asserted at write time**	The transmission-rate defect was caught by reading a run-log summary, not by the test suite. Bounded quantities — transmission ≤ 1, union $\geq$ both legs, intersection $\leq$ either — hold by construction and are checked ad hoc, if at all.	**Open.** Assert them where the values are written, so a units error fails the run rather than being noticed by whoever reads the log.

#	Landmine	Detail	Status
L10	**Silent-zero class of failure**	Records not corresponding to the storm catalogue produced a confident 0.0 conditional from 110 genuine flood events, with no error.	**Resolved for this path.** The conditional now raises on unresolved identifiers. The same class may exist elsewhere and has not been swept for.

12 Governance and registration

Per the new-model checklist, and noting that a new model document must be wired into **five registries** plus the Makefile filter:

1. Inventory entry `MKM-EF-001` — tier 1 (proposed), category Hazard, source module `src/models/frequency/`, upstream MKM-GH-001, downstream MKM-PR-001.
2. Corresponding upstream and downstream amendments on MKM-GH-001 and MKM-PR-001; the chain is hand-maintained and scanned by full-audit section 4.7.
3. `docs/models/event_frequency/event_frequency.tex` and Makefile, with the standard sections and the generated test-results and sensitivity inputs.
4. Registry wiring: five registries plus the Makefile filter.
5. Assumptions and limitations recorded — including, explicitly, the provenance limitation of Section 6 and the stationarity assumption.

Validation plan (SR 11-7). Conceptual soundness; per-gauge count backtesting on a held-out split; dispersion coverage reporting (share of gauges selecting NegBin); threshold sensitivity ($\hat{\lambda}$ and AAL under a one-band threshold perturbation); reconciliation of analytical against empirical AAL; the POT round-trip; annual recalibration with a drift alert; and an effective-challenge artefact emitted on every run.

Note that benchmarking return-period levels against published national flood-frequency estimates — version 1's validation test 6 — is **not meaningful** while catchments are synthetic. It is retained in the plan but flagged as pending real gauge data.

13 Risks and assumptions

Risk or assumption	Impact	Mitigation
λ has no observational content on synthetic catchments	Pricing is precise but not accurate; validation claims could overstate	Provenance class recorded on every fitted rate; round-trip framed as extraction validation, not calibration; real-data path built and ready

Risk or assumption	Impact	Mitigation
Repricing of 2–4× on every quote	Client and book impact	Stage 3 parallel-run report before switchover; staged rollout; legacy metric retained behind a flag
Event-definition change reprices before λ does	Two effects confounded in the impact analysis	Sequenced deliberately: Stage 1 isolates the denominator effect, Stage 3 the λ effect
Threshold choice materially moves $\hat{\lambda}$	Pricing instability	Sensitivity test; threshold governance in configuration with a change log
Overdispersion ignored where present	Understated tail	NegBin selected wherever the dispersion test rejects Poisson
Stationarity assumed	Understated forward risk under climate change	Explicit limitation in the model documentation; drift monitoring; Track B roadmap
Wind leg left on the legacy metric	Internally inconsistent peril comparison	Decide explicitly at Stage 4; do not leave implicit
Clamp removal widens the tail	Compounds the repricing impact	Quantify jointly with the repricing report, not separately

14 Glossary

AAL
: Average Annual Loss; expected loss per year.

AEP
: Aggregate Exceedance Probability; the probability that total annual loss exceeds x. *Track B.*

OEP
: Occurrence Exceedance Probability; the probability that the largest single event loss in a year exceeds x. *Track B.*

EEF
: Event Exceedance Frequency; the annual frequency, not probability, of events exceeding a level.

ELT / YLT
: Event Loss Table and Year Loss Table. *Track B.*

Event
: Here, the insurance hours-clause event — a storm *sequence* within the configured event window — not an individual storm.

POT
: Peaks Over Threshold; extreme-value sampling of all independent exceedances above a threshold.

GPD
: Generalised Pareto Distribution; the standard model for peaks-over-threshold exceedance magnitudes.

Dispersion index
: Variance divided by mean of annual counts; near one supports Poisson, above one indicates overdispersion.

NHPP
: Non-Homogeneous Poisson Process; a time-varying arrival rate. *Track B.*

Provenance class
: `observed`, `generator-derived` or `regional-fallback`; records what a fitted λ is actually based on.